Questions and MCQ

Step 3 : Tools : Cost of debt, Cost of equity and cost of capital

Use the following information to answer the question(s) below.

	Beta	Volatility
"Eenie"	0.45	20%
"Meenie"	0.75	18%
"Miney"	1.05	35%
"Moe"	1.20	25%

Assume that the risk-free rate of interest is 3% and you estimate the market's expected return to be 9%.

1) Which firm has the most total risk?
A) Eenie
B) Meenie
C) Miney
D) Moe
Answer: C
Explanation: Total risk is measured using volatility and Miney has the highest volatility, hence the most total risk.

Use the following information to answer the question(s) below.

Suppose the market consists only of Merck (MRK) and Boeing (BA). Merck stock is trading for $36.70 per share with 2.11 billion shares outstanding while Boeing has 697.5 million shares outstanding and a market capitalization of $38.223 billion. Assume that you hold the market portfolio.

2) Boeing's stock price is closest to:
A) $18.25.
B) $36.70.
C) $54.80.
D) $63.40.
Answer: C
Explanation: $\text{Price}_{BA} = \frac{\text{Market Cap}}{\text{Shares Out}} = \frac{\$38{,}223\ M}{697.5\ M} = \$54.80$

3) Merck's market capitalization is closest to:
A) $38.2 billion.
B) $77.4 billion.
C) $89.4 billion.

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D) $115.6 billion.
Answer: B
Explanation: Market Cap = Price × shares outstanding = $36.70 × 2,110 million = $77,437 million

4) Which of the following statements is FALSE?
A) Because very little trading is required to maintain it, an equal-weighted portfolio is called a passive portfolio.
B) If the number of shares in a value-weighted portfolio does not change, but only the prices change, the portfolio will remain value-weighted.
C) The CAPM says that individual investors should hold the market portfolio, a value-weighted portfolio of all risky securities in the market.
D) A price-weighted portfolio holds an equal number of shares of each stock, independent of their size.
Answer: A
Explanation: Because very little trading is required to maintain it, a value-weighted portfolio is called a passive portfolio.

5) In practice which market index is most widely used as a proxy for the market portfolio in the CAPM?
A) Dow Jones Industrial Average
B) Wilshire 5000
C) S&P 500
D) U.S. Treasury Bill
Answer: C

Use the following information to answer the question(s) below.

Year	Risk-free Return	Market Return	Wyatt Oil Return	Market Excess Return	Wyatt Oil Excess Return	Beta
2007	3.0%	6.0%	5.5%	3.0%	2.5%	0.833
2008	1.5%	-38.5%	-32.6%	-40.0%	-34.1%	0.853
2009	1.0%	22.5%	19.6%	21.5%	18.6%	0.865

6) Wyatt Oil's average historical return is closest to:
A) -2.50%.
B) -3.33%.
C) -4.33%.
D) -5.17%.
Answer: A
Explanation: $r_{average} = \frac{r_{2007} + r_{2008} + r_{2009}}{3}$

Year	Risk-free Return	Market Return	Wyatt Oil Return	Market Excess	Wyatt Oil Excess

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				Return	Return
2007	3.0%	6.0%	5.5%	3.0%	2.5%
2008	1.5%	-38.5%	-32.6%	-40.0%	-34.1%
2009	1.0%	22.5%	19.6%	21.5%	18.6%
Average	1.83%	-3.33%	-2.50%	-5.17%	-4.33%

7) The Market's average historical return is closest to:
A) -2.50%.
B) -3.33%.
C) -4.33%.
D) -5.17%.
Answer: B

Explanation: $r_{average} = \frac{r_{2007} + r_{2008} + r_{2009}}{3}$

Year	Risk-free Return	Market Return	Wyatt Oil Return	Market Excess Return	Wyatt Oil Excess Return
2007	3.0%	6.0%	5.5%	3.0%	2.5%
2008	1.5%	-38.5%	-32.6%	-40.0%	-34.1%
2009	1.0%	22.5%	19.6%	21.5%	18.6%
Average	1.83%	-3.33%	-2.50%	-5.17%	-4.33%

8) Wyatt Oil's average historical excess return is closest to:
A) -2.50%.
B) -3.33%.
C) -4.33%.
D) -5.17%.
Answer: C

Explanation: Excess $return_{average} = \frac{(r_{WO} - r_{rf})_{2007} + (r_{WO} - r_{rf})_{2008} + (r_{WO} - r_{rf})_{2009}}{3}$

Year	Risk-free Return	Market Return	Wyatt Oil Return	Market Excess Return	Wyatt Oil Excess Return
2007	3.0%	6.0%	5.5%	3.0%	2.5%
2008	1.5%	-38.5%	-32.6%	-40.0%	-34.1%
2009	1.0%	22.5%	19.6%	21.5%	18.6%
Average	1.83%	-3.33%	-2.50%	-5.17%	-4.33%

9) Which of the following statements is FALSE?
A) One difficulty when trying to estimate beta for a security is that beta depends on the correlation and volatilities of the future returns of the security and the market.
B) It is common practice to estimate beta based on the expectations of future correlations and volatilities.

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C) One difficulty when trying to estimate beta for a security is that beta depends on investors' expectations of the correlation and volatilities of the security's and market's returns.
D) Securities that tend to move less than the market have betas below 1.
Answer: B
Explanation: Beta is measured using past information.

Use the equation for the question(s) below.

Consider the following linear regression model:

$(Ri - rf) = ai + bi(RMkt - rf) + ei$

10) The *bi* in the regression:
A) measures the sensitivity of the security to market risk.
B) measures the historical performance of the security relative to the expected return predicted by the SML.
C) measures the deviation from the best-fitting line and is zero on average.
D) measures the diversifiable risk in returns.
Answer: A

Use the following information to answer the question(s) below.

Consider the following information regarding corporate bonds:

Rating	AAA	AA	A	BBB	BB	B	CCC
Average Default Rate	0.0%	0.1%	0.2%	0.5%	2.2%	5.5%	12.2%
Recession Default Rate	0.0%	1.0%	3.0%	3.0%	8.0%	16.0%	48.0%
Average Beta	0.05	0.05	0.05	0.10	0.17	0.26	0.31

11) Wyatt Oil has a bond issue outstanding with seven years to maturity, a yield to maturity of 7.0%, and a BBB rating. The corresponding risk-free rate is 3% and the market risk premium is 5%. Assuming a normal economy, the expected return on Wyatt Oil's debt is closest to:
A) 3.0%.
B) 3.5%.
C) 4.9%.
D) 5.5%.
Answer: B
Explanation: $r_d = r_{rf} + \beta(r_m - r_{rf}) = 3\% + 0.1(5\%) = 3.5\%$

12) Wyatt Oil has a bond issue outstanding with seven years to maturity, a yield to maturity of 7.0%, and a BBB rating. The bondholders' expected loss rate in the event of default is 70%. Assuming a normal economy the expected return on Wyatt Oil's debt is closest to:
A) 3.0%.

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B) 3.5%.
C) 4.9%.
D) 6.7%.
Answer: D
Explanation: r_d = YTM - prob(default) × loss rate = 7% - 0.5%(70%) = 6.65%

Use the following information to answer the question(s) below.

Consider the following information regarding corporate bonds:

Rating	AAA	AA	A	BBB	BB	B	CCC
Average Default Rate	0.0%	0.1%	0.2%	0.45%	2.2%	5.5%	12.2%
Recession Default Rate	0.0%	1.0%	3.0%	3.0%	8.0%	16.0%	48.0%
Average Beta	0.05	0.05	0.05	0.10	0.17	0.26	0.31

Company	Market Capitalization ($mm)	Total Enterprise Value ($mm)	Equity Beta	Debt Rating
Taggart Transcontinental	4500	8000	1.1	BBB
Rearden Metal	3800	7200	1.3	AAA
Wyatt Oil	2400	3800	0.9	A
Nielson Motors	1500	4400	1.75	BB

13) Your estimate of the debt beta for Taggart Transcontinental would be:
A) 0.05.
B) 0.10.
C) 0.17.
D) 1.00.
Answer: B
Explanation: Since Taggart has a rating of BBB, the appropriate debt beta from the table is 0.10.

14) Your estimate of the debt beta for Nielson Motors would be:
A) 0.10.
B) 0.17.
C) 1.00.
D) 1.68.
Answer: B
Explanation: Since Nielson has a rating of BB, the appropriate debt beta from the table is 0.17.

Use the following information to answer the question(s) below.

Consider the following information regarding corporate bonds:

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Rating	AAA	AA	A	BBB	BB	B	CCC
Average Default Rate	0.0%	0.1%	0.2%	0.5%	2.2%	5.5%	12.2%
Recession Default Rate	0.0%	1.0%	3.0%	3.0%	8.0%	16.0%	48.0%
Average Beta	0.05	0.05	0.05	0.10	0.17	0.26	0.31

15) Galt Industries has a market capitalization of $50 billion, $30 billion in BBB rated debt, and $8 billion in cash. If Galt's equity beta is 1.15, then Galt's underlying asset beta is closest to:
A) 0.83.
B) 0.92.
C) 1.00.
D) 1.15.
Answer: A
Explanation: We can think of Galt's business assets as a portfolio of equity, plus debt, and less cash. Assuming the beta of cash investments is zero:

$$\beta U = \frac{E}{E + D - C}\beta E + \frac{D}{E + D - C}\beta D + \frac{-C}{E + D - C}\beta C$$

$$= \frac{50}{50 + 30 - 8} \times 1.15 + \frac{30}{50 + 30 - 8} \times 0.10 + \frac{-8}{50 + 30 - 8} \times 0.0$$

$$= 0.84$$

Alternatively, we estimate Galt's asset beta based on its net debt of 30 - 8 = 22 m. Using net debt:

$$\beta U = \frac{E}{E + D - C}\beta E + \frac{\text{Net Debt}}{E + D - C}\beta D$$

$$= \frac{50}{50 + 30 - 8} \times 1.15 + \frac{30 - 8}{50 + 30 - 8} \times 0.10$$

$$= 0.829$$

Note that both answers are quite similar. The second approach presumes that Galt's cash reduces the average market risk of its debt (as though Galt used its cash to repay its senior debt).

16) Your firm is planning to invest in a new electrostatic power generation system. Electrostat Inc is a firm that specializes in this business. Electrostat has a stock price of $25 per share with 16 million shares outstanding. Electrostat's equity beta is 1.18. It also has $220 million in debt outstanding with a debt beta of 0.08. Your estimate of the asset beta for electrostatic power generators is closest to:
A) 0.76.
B) 0.79.
C) 0.93.
D) 1.10.
Answer: B
Explanation: $\beta U = \frac{E}{E + D}\beta E + \frac{D}{E + D}\beta D = \frac{\$25 \times 16}{\$25 \times 16 + 220} \times 1.18 + \frac{220}{\$25 \times 16 + 220} \times 0.08 = 0.789677$

17) If a project has a higher proportion of fixed to variable costs, holding the risk of its revenues constant:
A) its beta will be lower, hence its cost of capital will be lower.
B) its beta will be higher, hence its cost of capital will be higher.
C) its beta will be unaffected, since beta does not measure the sensitivity of the project's cash flows to

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market risk.
D) its financial leverage will be higher.
Answer: B

18) In a world with taxes, which of the following is the rate we should use to evaluate an all-equity financed project with the same risk as the firm?
A) The weighted-average cost of capital
B) The pre-tax WACC
C) The cost of equity
D) The cost of debt
Answer: B

19) Which of the following statements is FALSE?
A) We should be suspicious of beta estimates that are extreme relative to industry norms.
B) When using historical data, there is always the possibility of estimation error.
C) Evidence suggests that betas tend to revert toward zero over time.
D) For stocks, common practice is to use at least two years of weekly return data or five years of monthly return data when estimating beta.
Answer: C
Explanation: Evidence suggests that betas tend to regress toward the average beta of 1.0 over time.

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